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Whitepaper

AI and ESG:
How Generative AI is Transforming ESG Reporting

Table of Contents

Executive Summary 2

Growing Importance and Data Demands of ESG 3

State of ESG Reporting: What are the Current Issues? 4

How Artificial Intelligence Transforms ESG Reporting 7

Challenges with AI in ESG Reporting 10

Navigating AI-Driven ESG Reporting with Presgo 12

Executive Summary

AI tools are helping companies handle the growing complexity of ESG data and reporting. AI automates gathering and analysing large datasets, producing real-time insights and predictive analysis for better decision-making. However, AI also poses a threat to data security and privacy, as well as biases and discrimination in data assessment. These challenges include the risk of inaccurate data interpretation and reinforcing existing societal biases in ESG assessments. Despite these issues, about a third of companies plan to adopt digital technologies, including AI, to support their sustainability efforts in the coming years. Along with the evolution of technology, especially with generative AI being relatively new, there are new ways to stay ahead of potential vulnerabilities and enhance the opportunities it can provide.

Growing Importance and Data Demands of ESG

There is an increasing market demand for ESG-focused companies with accessible data on ESG initiatives. As consumers favour companies with a positive impact on the environment and society, investors also seek investment options that align with and contribute to sustainable developments and social equity.

This rising demand shows the importance of integrating ESG into corporate strategies. The 2024 state of corporate ESG report highlights that 82% of executives recognise how crucial the role of ESG is for long-term success. This pushes for better integration of ESG reporting into existing systems to navigate the complexities of meeting ESG standards. However, only 22% of CFOs feel fully prepared for the climate reporting and assurance requirements, signalling a need for more commitment and investment in ESG systems and compliance.

As sustainability standards continue to be adopted, they are also set to reshape industries and create changes in the job market. According to the World Economic Forum's data on the future of jobs, over 85% of organisations foresee digital innovations as a primary driver of transformation in the next five years.

Among the most impactful trends are the integration of ESG standards and the green transition, which are expected to generate job growth, particularly in fields like AI, sustainability, and climate management. Technologies such as big data, cloud computing, and AI are expected to be widely adopted, with 75% of companies planning to integrate them into their operations by 2027. This shift anticipates new job opportunities, including roles for AI machine learning, which was the number one in the top highest-growing roles, followed by sustainability experts.

State of ESG Reporting: What are the Current Issues?

Data Volume and Complexity

The importance of ESG data has surged over the past decade, with over 30 countries now requiring ESG disclosures. According to 2024 KPMG data, 95% of the world's largest 250 companies are now setting carbon reduction targets. An increase is evident from 2022's 80%, and over half of these companies have appointed a sustainability leader.

Despite these efforts, organisations face challenges in collecting accurate and reliable data. Companies are not only overwhelmed by the sheer volume of ESG data required for disclosure, but also by the multiple standards and regulatory proposals they must comply with. The inconsistency of data requests pressures companies to sift through large amounts of data. This makes it difficult to focus on actionable insights.

Poor-quality ESG data was identified as the biggest barrier to adopting sustainable investing. In another report, 71% of investors pointed out incomplete and inconsistent ESG data as a major obstacle, for which respondents in the Asia-Pacific (APAC) have almost a similar percentage with the same sentiment. They expected it would improve over time, but the issue persisted as observed in earlier surveys. The subjective, incomplete, or inconsistent nature of ESG data makes it difficult to assess performance or manage risk effectively.

State of ESG Reporting: What are the Current Issues?

Evolving Regulatory Requirements

ESG reporting regulations are evolving rapidly, with stricter rules on accountability and transparency on ESG practices. This pushes companies to adopt and improve transparent and efficient methods of analysing and reporting their ESG data. For instance, the European Union's Corporate Sustainability Reporting Directive (CSRD) now requires companies with more than 500 employees to disclose detailed ESG data. It is also for expansion to SMEs and non-EU companies with EU operations. In addition, the EU introduced an omnibus proposal that will consolidate and enhance the reporting guidelines of different directives.

Similar trends are taking shape in the Asia-Pacific region, where countries are advancing their sustainability practices to align with global standards. Singapore is leading with regulations and initiatives, creating the Singapore Sustainable Finance Association (SSFA) to boost collaboration in the financial sector in tackling the net-zero financing gap. The country's Accounting and Corporate Regulatory Authority (ACRA) and the Singapore Exchange (SGX) established the Sustainability Reporting Advisory Committee (SRAC) to improve corporate sustainability disclosures.

In Hong Kong, new ESG regulations from the Hong Kong Exchanges and Clearing Limited (HKEX) will start to require Scopes 1 and 2 emissions reporting, with Scope 3 advised, beginning in 2025. On top of that, China has introduced its first Corporate Sustainability Disclosure Guidelines, aiming for full implementation by 2030, while working with the EU on a shared green investment framework to align their sustainability efforts across various sectors.

State of ESG Reporting: What are the Current Issues?

ESG-Based Decision-Making

CEOs recognise the threats of climate change and technological disruption to their companies. In a survey, 45% of CEOs worry that their company may not survive within the next decade without significant transformation. While many are taking steps toward reinvention, concerns about long-term survival remain high. Technological developments and climate change are two major forces driving these shifts, and many are adjusting their business models to incorporate climate-friendly investments and sustainability. This is also influenced by the rising investor interest in sustainable investments at 77%, globally.

Strong ESG practices are seen as capable of delivering high returns, yet more than half of investors feel they still lack proper financial advice and clear sustainable investment options. This uncertainty creates a barrier to effective decision-making and emphasises the need for clearer, more reliable, and transparent ESG reporting to help investors make informed choices.

How Artificial Intelligence Transforms ESG Reporting

As AI continues to be integrated into business strategies, companies are seeing tangible improvements in advancing various aspects of the reporting process. Leaders in companies such as those in the retail sector achieve three times better performance, compared to their competitors. Over half of CEOs report that AI has already increased employee efficiency. Through generative AI and machine learning, businesses can reduce errors, improve accuracy, and guarantee compliance with the rapidly changing sustainability regulations.

Data Collection and Analysis

AI has a strong potential to automate the collection and analysis of diverse data from multiple sources, including financial reports, satellite imagery, and public records. This automation streamlines the data collection process, making sure businesses can gather and analyse comprehensive ESG data quickly and accurately.

By utilising most of the tools driven by ESG and AI, companies can process vast amounts of unstructured data from multiple sources. This allows them to extract actionable insights to improve decision-making and sustainability strategies.

Advanced technology, such as OpenSC, utilises AI, blockchain, IoT sensors, and machine learning to help companies with data collection and analysis, traceability, and verification of supply chains. A collaboration with the World Wildlife Fund (WWF), the platform focuses on sustainability, particularly in ethical food production. It helps companies verify claims on their resources and business practices and allows consumers to trace products from their source. Digital technologies, such as AI, IoT, and blockchain, are expected to not only refine and simplify reporting but also cover broader environmental sustainability and make data-driven adjustments to their ESG strategies efficiently.

The reliance on third-party AI and ESG tools is growing, with 97% of companies using external solutions to manage ESG activities. These technologies are highly essential in data automation and collection, improved accurate reporting, and assured compliance with evolving regulations. AI's predictive capabilities also support better decision-making and strategy development, helping companies stay ahead in the playing field within regulatory changes and improve sustainability efforts.

How Artificial Intelligence Transforms ESG Reporting

ESG Performance Analysis and KPI Tracking

ESG machine learning algorithms help businesses identify significant trends and anomalies within ESG data. This capability provides insights into long-term performance patterns and tracking sustainability reporting (SR) scoring, allowing companies to recognise areas of improvement or emerging risks that could impact their sustainability efforts. AI's ability to analyse large datasets quickly and accurately allows businesses to track trends that would go unnoticed. By recognising these patterns, companies can better anticipate the future and proactively adjust their strategies to stay aligned with their sustainability goals.

A recent survey indicated that AI is impactful in reshaping organisations' approach to ESG, focusing on identifying patterns and trends within ESG data. Key findings show that AI-driven systems greatly improve the ability to track ESG metrics and key performance indicators (KPIs) in real-time, offering businesses the ability to make quick adjustments. Predictive analysis further strengthens ESG data analysis by forecasting future sustainability risks and opportunities. This approach raises the accuracy of ESG reporting by helping companies predict ESG scores with near-perfect accuracy. This was the case in a recent study involving Fortune 500 companies, which had observed 99% accuracy in predicting performance scores.

AI tools for ESG, including machine learning algorithms, help organisations identify performance patterns and trends in ESG data. These insights help businesses anticipate sustainability challenges and risks, making sure they stay ahead of regulatory requirements and ESG KPIs. Companies using AI are also more likely to integrate risk management strategies into their ESG frameworks, with greater focus on areas such as cybersecurity, data governance, and compliance risks. Shortly, more companies are expected to rely on AI to analyse data and derive insights for sustainability decision-making and investing.

How Artificial Intelligence Transforms ESG Reporting

Regulatory, Climate, and Security Risks Management

A PwC survey of 325 global investors found that 79% view ESG risks as crucial in investment decisions, and 68% believe executive compensation should be linked to ESG performance. More companies are reporting their climate-related risk management process, with 81% in 2024. This is higher than 74% in the previous year.

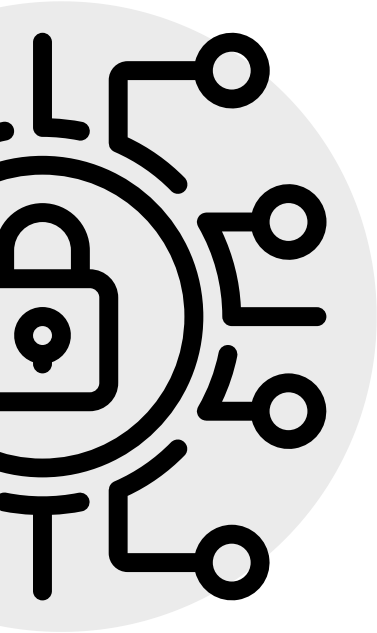
Companies are relying so much more on AI tools to make sure they effectively assess and manage regulatory, climate, and security risks. AI is a powerful resource, helping organisations navigate regulations and identify compliance gaps by creating tailored ESG-compliant reports, comparing policies, and identifying gaps. It can automate the compliance-checking process, alerting teams to potential breaches, misalignments, and early warning signals. It also monitors climate-related risks and cybersecurity threats, providing predictive insights into potential vulnerabilities. This level of automation guarantees that ESG reports are generated on time and meet the required regulatory criteria. This is while integrating risk management in various domains to speed up the process and ensure findings are up-to-date and to the highest standard for ESG strategies.

Report Generation

Advancements in AI, particularly generative AI, will have a substantial part in transforming ESG report generation. AI tools are now capable of efficient data collection, analysis, and processing, reducing a large amount of workload and human error in report generation. Integrating AI into ESG tools, this technology is expected to become more sophisticated, from extracting data from unstructured sources to generating optimised and thoroughly translated disclosures. This will lead to specialised solutions and insights, offering a qualitative narrative of the SR based on quantitative data. As they continue to evolve, investments in AI are also foreseen to increase by both established and upcoming players to further automate the creation of sustainability disclosures.

Challenges with AI in ESG Reporting

While AI offers massive advantages for ESG reporting, its integration into sustainability practices comes with challenges. These are primarily centered around data privacy, security, bias, and skills gap. Addressing these issues can fully realise the potential of AI-driven tools.



Data Privacy and Security

AI integration into ESG reporting opens many benefits but also introduces major risks, particularly around data privacy and security. AI systems require access to massive amounts of sensitive and confidential data. This increases the vulnerability to breaches and unauthorised access, which undermines the credibility and trust in AI-driven ESG reports. In a recent 2025 case, for example, a ransomware group compromised Schneider Electric's cloud-based platform that gathers sustainability data to track and predict emissions for reporting. With large amounts of sensitive information that can be stolen and manipulated, cyber attacks like this can damage the integrity of ESG data.

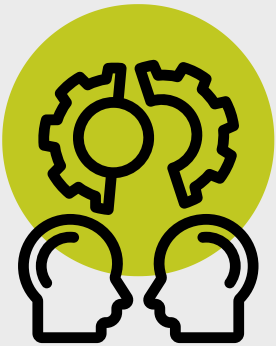
Adopting cybersecurity policies and secure ESG software to handle this data not only protects against external threats but also secures data privacy regulations like the General Data Protection Regulation. This can be observed in companies prioritising data protection and cloud security in 2025, with 48% of business executives placing the highest importance on data protection and 34% focusing on cloud security.

Challenges with AI in ESG Reporting

Dealing with Bias and Discrimination

AI systems sometimes reflect societal biases present in the training data or are created by developers that may not be fully representative of the bigger group. A lack of diversity in the research and development teams increases the likelihood that cultural, gender, and racial biases will influence AI models. This bias in AI models can lead to unequal treatment of assessment and data.

Research identified that improving training data by applying stratified sampling and adversarial data can keep a check on the fairness and effectiveness of AI models. With fairness-aware algorithms, AI can focus on achieving more equitable outcomes, help avoid bias issues, and improve diversity and overall performance.



Training and Skills Gap

Despite how organisations see AI driving high turnover, a significant skills gap still exists, as 60% of workers will need training by 2027, with analytical thinking and AI expertise being top priorities for upskilling. With this concern, about 84% of employees from countries in Australia, India, Singapore, New Zealand, and the UK are substantially supported by their companies in their AI skills development. This response is compared to only 50% for employees in the U.S., even though they hold similar opinions about AI adoption. This global skills gap and the need for more training need to be addressed. Companies must prioritise the continuous upskilling of employees moving forward to effectively maximise AI's potential and continuous developments.

Navigating AI-Driven ESG Reporting with Presgo

As AI transforms business operations, it presents substantial opportunities for improving ESG reporting. The reporting process requires software solutions designed to address the unique challenges of sustainability tracking. It heavily benefits from utilising AI to simplify, speed up, and elevate the process for a more reliable and heavily analysed disclosure. All these are possible with advanced ESG reporting software, equipped with AI and capable of handling the complexity and volume of ESG data.

Presgo, a leading ESG reporting software, provides a flexible, AI-powered platform that adapts to industry-specific needs. It offers AI assistance and AI-powered capabilities, especially data analysis, tracking, translation, and generating fully assessed and monitored disclosures.

Here is how Presgo can transform your ESG reporting

Improved efficiency in reporting

Automation of data collection and analysis, **AI-assisted translations,** and **AI-powered ESG support** result in simplified workflows and efficient reporting and compliance.

Improved accuracy and transparency

Validation tools, including its **Optical Character Recognition (OCR)** technology, process and capture large datasets more accurately. This feature detects anomalies and helps companies deliver more reliable and transparent ESG reports.

Scalable integration of ESG Metrics

It is equipped with **AI-assisted SR scoring**, aligning with global frameworks such as **GRI and TCFD**. Its goal-setting and tracking features incorporate ESG metrics and benchmarking into daily operations. This integration makes sustainability a core part of the business performance and ESG tracking.

Real-time monitoring and adaptation

Advanced systems continuously monitor ESG performance, providing companies with real-time insights and analysing trends for quicker adaptation to regulatory changes and decision-making.

By embracing this technology, companies are guaranteed an ESG report of the highest calibre, a transformative step toward sustainability.

See how Presgo can empower your ESG reporting with AI.

Book a Demo Now

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